Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	5 April 2026
Team ID	SWTID-2026-5153
Project Name	Automating employee data
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement	Sub Requirement
FR-1	Employee Data Import	Upload employee data through CSV/XLSX file Automatic creation of import table (import_employee)Validate uploaded file format and data
FR-2	Data Transformation & Mapping	Create Transform Map between import_employee and sys_user Map basic fields (Name, Email, Phone) Map reference fields (Department, Manager) using Mapping Assist
FR-3	Data Validation & Processing	Execute Transform Map to insert records. Validate correct data insertion into sys_user table. Log errors for invalid or missing data
FR-4	Incident Form Automation	Display employee details on Incident form. Show Department using dot-walking. Show Manager details (Email) using dot-walking.
FR-5	Dynamic Data Display	Auto-update fields when "Assigned to" user changes No page reload required for data update
FR-6	Security & Access Control	Restrict data import access to authorized users Ensure data visibility based on roles (ACLs)

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The system should provide a user-friendly interface for uploading employee data and viewing details. The Incident form should automatically display employee information without requiring extra steps or navigation.
NFR-2	**Security**	Only authorized users (Admin/Import roles) should be allowed to upload and manage employee data. Data visibility must follow Role-Based Access Control (RBAC) and Access Control Lists (ACLs).

NFR-3	**Reliability**	The system should handle incorrect or missing data gracefully. Errors during data import should be logged without stopping the entire process.
NFR-4	**Performance**	The system should process data imports quickly without affecting pla form should load and display data instantly without delay.
NFR-5	**Availability**	The system should be available at all times with minimal downtime. Data import and Incident form functionalities should work consistently.
NFR-6	**Scalability**	The system should support both small and large datasets (e.g., thousands of employee records) without performance degradation. It should scale efficiently as data increases.